

**Alg2 Precalc Acc.
Sem 2 Final Review (sem 1
content)**

Chapter 1 - Sets/Inequalities/Abs Val/Piecewise

Sets + Notation

Sets: A collection of objects known as elements of the set

Set Notation

- Sets use curly braces $\rightarrow \{ \}$
 - $\in$ means “is an element of”
 - $\notin$ means “is NOT an element of”

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$a \in S$$

$$b \notin S$$

Sets + Notation (Continued)

Suppose we have n sets $\rightarrow S_1, S_2, S_3 \dots S_n$

Union: **ALL** elements that are in S_1, S_2, S_3 , all the way to S_n

- Use the symbol **U** to define the **UNION** of sets

Intersection: **ALL** elements that S_1, S_2, S_3 , all the way to S_n **SHARE IN COMMON**

- Use the symbol **∩** to define the **INTERSECTION** of sets

$$S_1 \cup S_2 \cup S_3 \cup \dots \cup S_n$$

$$S_1 \cap S_2 \cap S_3 \cap \dots \cap S_n$$

$$\bigcup_{i=1}^n S_i$$

$$\bigcap_{i=1}^n S_i$$

Intervals + Notation

Intervals: Sets of **REAL** numbers between 2 fixed endpoints

Open Intervals: All real numbers between 2 endpoints

EXCLUDING the endpoint(s)

- Denoted using parenthesis $\rightarrow ()$
- When an endpoint is $+\infty$ or $-\infty$, the endpoint is **ALWAYS** open
- **UNFILLED** circles on a number line

Closed Intervals: All real numbers between 2 endpoints **INCLUDING** the endpoint(s)

- Denoted using square brackets $\rightarrow []$
- **FILLED** circles on a number line

Notation	Set description	Graph
(a, b)	$\{x \mid a < x < b\}$	
$[a, b]$	$\{x \mid a \leq x \leq b\}$	
$[a, b)$	$\{x \mid a \leq x < b\}$	
$(a, b]$	$\{x \mid a < x \leq b\}$	
(a, ∞)	$\{x \mid a < x\}$	
$[a, \infty)$	$\{x \mid a \leq x\}$	
$(-\infty, b)$	$\{x \mid x < b\}$	
$(-\infty, b]$	$\{x \mid x \leq b\}$	
$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	

Practice (Very Easy)

Write the following sets as a single set, and graph on a number line

$$(1, 3) \cap [2, 7]$$

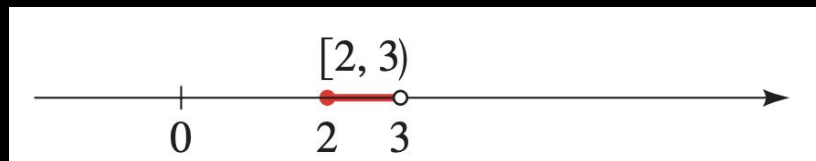
$$(1, 3) \cup [2, 7]$$

Answers

$$(1, 3) \cap [2, 7]$$



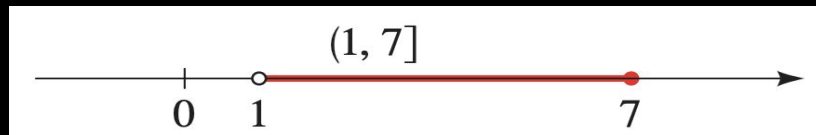
$$[2, 3)$$



$$(1, 3) \cup [2, 7]$$



$$(1, 7]$$



How to Solve Inequalities

Step 1: Get all the terms to one side (preferably left)

Step 2: Write as a **SINGLE** fraction and get 0 on the other side

Step 3: Factorize the numerator/denominator as much as possible

Step 4: Take each factorized term and set it equal to 0, then solve for the variable. These are our **CRITICAL**

POINTS

Step 5: Draw a number line with the critical points, and determine the sign of the function in each interval

Step 6: Use the signs to find the solutions to the inequality

NOTE: Write solutions in interval notation with **U**, representing the **UNION** of all solutions, and consider the domain when deciding open/closed inequalities for inequalities with $\geq$ or $\leq$ symbols

Practice (Fairly Hard)

Solve the following inequality

$$\frac{4x}{x^3 + 1} \geq \frac{2x + 8}{x + 1} - 3$$

Solution

$$(-\infty, -1) \cup \left[1, \frac{5 - \sqrt{5}}{2}\right] \cup \left[\frac{5 + \sqrt{5}}{2}, \infty\right)$$

Absolute Value & Distance

- Absolute Value: Distance from 0
 - Absolute value is **ALWAYS** positive
 - Denoted with $| |$
- Distance **CANNOT** be negative
 - Imagine saying “I walked -2 miles today 🏴‍☠️”
- When the inside of an absolute value is zero,
NOT JUST THE VARIABLE, we call this a turning point
 - Set inside = 0 and solve for x, or whatever the variable in question is, let's say $x=a$.
 - $x > a \rightarrow +(\text{inside of abs val})$
 - $x < a \rightarrow -(\text{inside of abs val})$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$|x + 3| = \begin{cases} x + 3 & \text{if } x + 3 \geq 0 \\ -(x + 3) & \text{if } x + 3 < 0 \end{cases}$$

$$|x + 3| = \begin{cases} x + 3 & \text{if } x \geq -3 \\ -x - 3 & \text{if } x < -3 \end{cases}$$

Writing Absolute Value as a Piecewise Function

Step 1: Set the inside of the absolute value function to 0 and solve for the variable in question

Step 2: At the point where the variable is 0, determine the behavior of the absolute value term

- There can be many absolute value terms, look at it part by part

Step 3: Put it all together

How to determine behavior: When the variable is $\geq$ than the turning point, absolute value function behaves like $+(inside)$, vice versa

Practice (Easy-Medium)

Write as a piecewise

$$|2x - 5| + 3|x + 1| - 4$$

Composition of Functions

$$(f \circ g)(x) = f(g(x))$$

$$(g \circ f)(x) = g(f(x))$$

$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

$$\text{Domain}(f \circ g) = \{x | x \in D_g, g(x) \in D_f\}$$

$$\text{Domain}(g \circ f) = \{x | x \in D_f, f(x) \in D_g\}$$

$$D_{f+g} = D_f \cap D_g$$

$$D_{fg} = D_f \cap D_g$$

$$D_{f/g} = D_f \cap D_g, g(x) \neq 0$$

Practice (Easy-Medium)

State the domain of the composition, $(f \circ g)(x)$ and write as a single expression

$$\text{Let } f(x) = 2x + 3$$

$$g(x) = x^2$$

$$f(x) = \sqrt{x - 1}$$

$$g(x) = 3x + 2$$

Solutions

$$x \in \mathbb{R}$$

$$x \in \left[-\frac{1}{3}, \infty\right)$$

Practice (Extremely Hard)

State the domain of the composition, $(f \circ g)(x)$

$$f(x) = \frac{2}{x - 3}$$

$$g(x) = \frac{x + 1}{x^2 - 4}$$

Solution

$$(-\infty, -2) \cup (-2, 1 - \frac{\sqrt{157}}{6}) \cup (1 - \frac{\sqrt{157}}{6}, 2) \cup (2, 1 + \frac{\sqrt{157}}{6}) \cup (1 + \frac{\sqrt{157}}{6}, \infty)$$

Chapter 2 - Functions

Inverses

1. Identify the domain and range of the **ORIGINAL** function
 2. Range of the original function $\rightarrow$ domain of inverse and domain of original function $\rightarrow$ range of inverse
 3. If the function is of the form $y=\dots x$, then switch y and x , then solve for y
- Properties of inverses and original functions
 - If $g(x)$ is the inverse of $f(x)$, then $f(g(x))=x$ and $g(f(x))=x$
 - Functions only have a single inverse function

Practice (Easy)

Find the inverse of the function, and state the domain/range of the inverse

$$h(x) = \frac{1 + 2x}{7 + x}$$

Solution

$$h^{-1}(x) = \frac{7x - 1}{2 - x}$$

D: $x \neq 2$

R: $y \neq -7$

Review: How to Solve Inequalities

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NOTE: Write solutions in interval notation with **U**, representing the **UNION** of all solutions, and consider the domain when deciding open/closed inequalities for inequalities with $\geq$ or $\leq$ symbols

Practice (Easy-Medium)

Solve the following inequality

$$\frac{3}{x+2} \leq \frac{4}{x-9}$$

$$\frac{x^2 + 4x - 21}{x^2 - 6x + 9} > 0$$

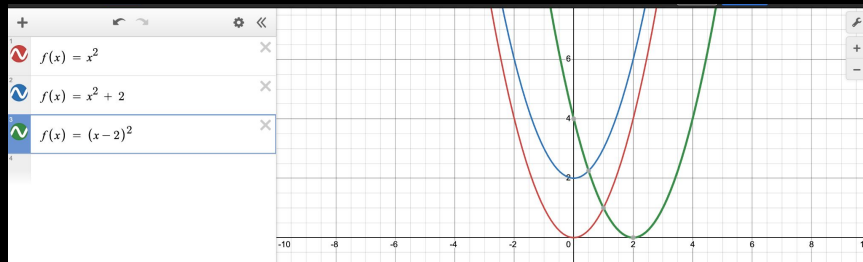
Solution

$$[-35, -2) \cup (9, \infty)$$

$$(-\infty, -7) \cup (3, \infty)$$

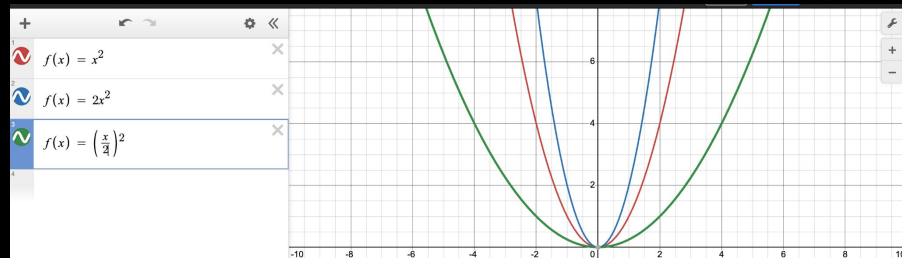
Transformations of Functions

- Translations:
 - Vertical translation: $(x,y) \rightarrow (x,y+k)$
 - $f(x) \rightarrow f(x) + k$
 - Horizontal translation: $(x,y) \rightarrow (x+k, y)$
 - $f(x) \rightarrow f(x-k)$
 - IMPORTANT: Horizontal transformations use the inverse operations of what they do to a function (minus sign for shifting right, etc.)



Transformations of Functions

- Stretches and Shrinks
 - Vertical:
 - $(x,y) \rightarrow (x,ky)$
 - $f(x) \rightarrow kf(x)$
 - Horizontal:
 - $(x,y) \rightarrow (kx,y)$
 - $f(x) \rightarrow f((1/k)x)$
 - IMPORTANT: Similar to horizontal transformations, $1/k$ is being substituted into the function instead of k in the horizontal stretch/shrink.



Order of Operations for Transformations

If the function is in $y=f(x)$ form:

- Work from outside to inside for vertical transformations applied on the entire function (not related to just an x-component).
- For horizontal transformations, write transformations to bring the function to the parent function as if you are solving for x.
- Write out the steps for transformations to get the new coordinates after the transformation from the parent function.



Example:

Parent function: $f(x) = x^2$

Transformations:

Horizontal: $x \rightarrow 2x + 2$

Vertical: $y \rightarrow 3y - 2$

Coordinates: $(x, y) \rightarrow (2x+2, 3y-2)$

Intermediate Value Theorem (IVT)

IVT: Existence theorem to prove the existence of a value on an interval, **DOESN'T** disprove the existence of roots

Let $f(x)$ be a function $\rightarrow$ There exists **AT LEAST 1** $c \in (a, b)$ and $f(c) = M$ between $f(a)$ and $f(b)$ if

1. $f(x)$ is continuous on the interval $[a, b]$
2. $f(a) \neq f(b)$

Practice (Medium)

Does there exist a c between 0 and 1 where $f(c)=\frac{1}{2}$?

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Solution

Yes

While $\sin(1/x)$ isn't continuous on $[0, 1]$, it is continuous on $[1/\pi, 1]$ as $\sin$ is a continuous function, but $1/x$ is discontinuous at $x=0$, so it is continuous on any interval not containing $x=0$

1. Continuous on $[1/\pi, 1]$
2. $f(1) = \sin(1/1) = \sin(1) \rightarrow$ in degrees this is just $\sin(180/\pi^\circ)$. While we don't know $\sin(180/\pi^\circ)$, we know that $180/\pi^\circ$ is greater than 30° , and $\sin(30^\circ)$ is $\frac{1}{2}$, and since $\sin$ is an increasing function from 0° to 90° , $\sin(180/\pi^\circ) > \frac{1}{2}$. 1 lies in the interval of continuity
3. Solving for the zeroes of $\sin(1/x)=0$, we get $1/x = \pi$ or $1/x = 2\pi \rightarrow x=1/\pi$ or $1/2\pi$, which lies in the interval of continuity established earlier

There indeed exists a c between $[0, 1]$ where $f(c)=\frac{1}{2}$, as $1/\pi \leq c \leq 1 \rightarrow [1/\pi, 1] \in [0, 1]$, and $0 \leq \frac{1}{2} \leq \sin(1)$

Finding the Domain of a Function

- Start from the outside function $\rightarrow$ find its domain
- Find the domain of each function which composes the larger function
- Take the **INTERSECTION** of each function's domain

Practice (Hard)

State the domain of the function

$$f(x) = \sqrt{\frac{x+1}{x-4\sqrt{x+4}}}$$

Solution

$$[-4, -1] \cup (8 + 8\sqrt{2}, \infty)$$

Chapter 4 - Trig

Graphing Trig Functions

The sine and cosine curves

$$y = a \sin k(x - b) \quad \text{and} \quad y = a \cos k(x - b) \quad (k > 0)$$

have **amplitude** $|a|$, **period** $2\pi/k$, and **horizontal shift** b .

An appropriate interval on which to graph one complete period is $[b, b + (2\pi/k)]$.

Graphing Trig Functions

$$y = -4 \cos \left(\frac{1}{2}x + \frac{\pi}{6} \right) - 2$$

$$y = -3 \sin \left(2x - \frac{\pi}{3} \right) + 4$$

Graphing Trig Functions

$$y = \sin x + \cos x$$

$$\sqrt{6} \sin \left(3x + \frac{4\pi}{3} \right) + \sqrt{2} \cos \left(3x + \frac{4\pi}{3} \right)$$

More Graphing

The functions tangent and cotangent have period π :

$$\tan(x + \pi) = \tan x \qquad \cot(x + \pi) = \cot x$$

The functions cosecant and secant have period 2π :

$$\csc(x + 2\pi) = \csc x \qquad \sec(x + 2\pi) = \sec x$$

More Graphing

1. Find the $t(x,y)$ for the function
2. Use the $t(x,y)$ to find all asymptotes (**know asymptotes for trig functions!!**)
3. Use the $t(x,y)$ to graph around 4-5 points
4. Sketch a **NICE** curve of the function

Notes:

- **LABEL ASYMPTOTES!!!**
- Show work for finding the new points (don't do mentally)

Graphing Trig Functions

$$y = \tan\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

$$y = -2 \cot(x) + 1$$

Solving Trig Equations

- Get **EVERYTHING** in terms of 1 function (all sine, cos, tan...)
- If you are solving a quadratic, let $u = \sin(x)$, $\cos(x)$, $\tan(x)$... (or any variable you like)
- Consider the domain of the functions ($\tan(x)$, $\csc(x)$...)
 - Also consider domain of the overall function
- ASTC!!
 - Q1: all are positive, Q2: sin is positive, Q3: tan is positive, Q4: cos is positive

Solving Trig Equations (Important!!)

- $\sin(x) = y$
 - $x = \arcsin(y)$ **AND** $\pi - \arcsin(y)$
- $\cos(x) = y$
 - $x = \arccos(y)$ **AND** $2\pi - \arccos(y)$

PLEASE KNOW THESE BY HEART

$$\cos \theta = \sin\left(\frac{\pi}{2} - \theta\right)$$

$$\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$$

$$\cot \theta = \tan\left(\frac{\pi}{2} - \theta\right)$$

$$\tan \theta = \cot\left(\frac{\pi}{2} - \theta\right)$$

$$\csc \theta = \sec\left(\frac{\pi}{2} - \theta\right)$$

$$\sec \theta = \csc\left(\frac{\pi}{2} - \theta\right)$$

Solving Trig Equations

$$\text{Solve for } x: (\csc x)(\cot x) = \sqrt{2}$$

$$\text{Solve for } x: \frac{\tan(x) - \cot(x)}{\tan^2(x) - \cot^2(x)} = \frac{1}{2}$$

Co-Function Fun

Simplify: $\arctan(x) + \operatorname{arccot}(x)$

$$\arctan(x) + \arctan\left(\frac{1}{x}\right) = \frac{\pi}{2}$$

DON'T GO TO YOUR FINAL W/O KNOWING THESE

Reciprocal Identities

$$\csc x = \frac{1}{\sin x} \quad \sec x = \frac{1}{\cos x} \quad \cot x = \frac{1}{\tan x}$$

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1 \quad \tan^2 x + 1 = \sec^2 x \quad 1 + \cot^2 x = \csc^2 x$$

Even-Odd Identities

$$\sin(-x) = -\sin x \quad \cos(-x) = \cos x \quad \tan(-x) = -\tan x$$

Cofunction Identities

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x \quad \tan\left(\frac{\pi}{2} - x\right) = \cot x \quad \sec\left(\frac{\pi}{2} - x\right) = \csc x$$

$$\cos\left(\frac{\pi}{2} - x\right) = \sin x \quad \cot\left(\frac{\pi}{2} - x\right) = \tan x \quad \csc\left(\frac{\pi}{2} - x\right) = \sec x$$

Basic Trig Formulas

Sum and Difference Identities

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Double Angle Identities

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= 2 \cos^2 \theta - 1$$

$$= 1 - 2 \sin^2 \theta$$

More Trig Formulas

Power-Reducing Formulas

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

Half-Angle Formulas

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\begin{aligned} \tan \frac{\theta}{2} &= \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \\ &= \frac{1 - \cos \theta}{\sin \theta} \\ &= \frac{\sin \theta}{1 + \cos \theta} \end{aligned}$$

Triple Angle Formulas

$$\sin 3\alpha = 3 \sin \alpha - 4 \sin^3 \alpha = 4 \sin \alpha \cdot \sin \left(\frac{\pi}{3} - \alpha \right) \cdot \sin \left(\frac{\pi}{3} + \alpha \right)$$

$$\cos 3\alpha = -3 \cos \alpha + 4 \cos^3 \alpha = 4 \cos \alpha \cdot \cos \left(\frac{\pi}{3} - \alpha \right) \cdot \cos \left(\frac{\pi}{3} + \alpha \right)$$

$$\tan 3\alpha = \frac{3 \tan \alpha - \tan^3 \alpha}{1 - 3 \tan^2 \alpha} = \tan \alpha \cdot \tan \left(\frac{\pi}{3} - \alpha \right) \cdot \tan \left(\frac{\pi}{3} + \alpha \right)$$

$$\cot 3\alpha = \frac{-3 \cot \alpha + \cot^3 \alpha}{-1 + 3 \cot^2 \alpha} = \cot \alpha \cdot \cot \left(\frac{\pi}{3} - \alpha \right) \cdot \cot \left(\frac{\pi}{3} + \alpha \right)$$

More Trig Formulas

Sum/Difference to Product Formulas

$$\sin \alpha + \sin \beta = 2 \sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2}$$

$$\sin \alpha - \sin \beta = 2 \sin \frac{\alpha - \beta}{2} \cos \frac{\alpha + \beta}{2}$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{\alpha - \beta}{2} \sin \frac{\alpha + \beta}{2}$$

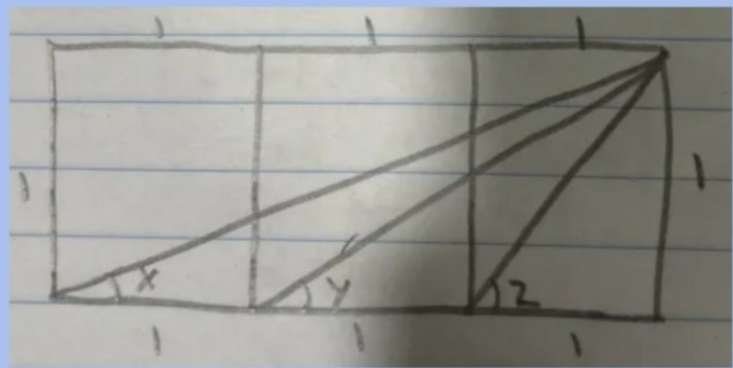
Product-to- Sum Formulas

$$\sin \alpha \cdot \cos \beta = \frac{1}{2} [\sin(\alpha - \beta) + \sin(\alpha + \beta)]$$

$$\sin \alpha \cdot \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cdot \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

Simplify: $\sin(2 \arctan(\sqrt{7 - 3x}))$



$$\tan(\operatorname{arccsc}(\frac{x}{4}))$$